

BAK ALI AREF

M2 Student in AI & Data Science | 6-Month Final-Year Internship – February 2027

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Summary

M2 student in Artificial Intelligence, Data Science and Agentic AI at Université Paris Dauphine-PSL, with hands-on experience in Machine Learning, Deep Learning, LLMs, AI agents, RAG, Computer Vision and MLOps, and in building end-to-end AI solutions. I am looking for a 6-month final-year internship starting February 15, 2027, in Artificial Intelligence, Data Science, Machine Learning or Agentic AI.

Skills

Languages: Python, R, Java, JavaScript, C, MATLAB, SQL.
AI & Data Science: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, Generative AI, AI Agents (LangGraph, LangChain), RAG (Retrieval-Augmented Generation), Classification, Regression, Clustering, Bayesian Networks.
Libraries: Scikit-learn, Pandas, NumPy, TensorFlow/Keras, PyTorch, Hugging Face (Transformers), XGBoost, OpenCV, MediaPipe, pyAgrum, SHAP, LIME, Geopandas.
MLOps & Tools: MLflow, Evidently, CI/CD (GitHub Actions), Docker, Git, vLLM, Streamlit, Google Cloud (Vertex AI).
Web & Databases: FastAPI, Flask, Django, React.js, MySQL, PostgreSQL.

Education

Université Paris Dauphine-PSL – Tunis Campus **09/2025 – 2027 (expected)**
Master's in Artificial Intelligence, Data Science and Agentic AI (M1 completed, M2 in progress)
Key courses: Machine Learning, Deep Learning, NLP, Bayesian Networks, Time Series

Université Paris Dauphine-PSL – Tunis Campus **09/2022 – 05/2025**
Bachelor's in Applied Mathematics

Lycée Aboulkacem Chebbi (La Chebba) – Baccalaureate in Technical Sciences, with Highest Honors **2021 – 2022**

Experience

AI Intern – Conversational Scheduling Agent **07/2026 – 09/2026**
ARSII – Remote Internship

- Designed and deployed a multi-tool conversational agent (**LangGraph**, *tool calling*): event extraction and conflict detection with **87% accuracy** on 60 test cases; open-source LLM served via **vLLM** (1.6 s average latency), FastAPI, React/TypeScript, Google Calendar/Gmail integration.

Intern – Educational AI Agent (Math+ Program) **07/2026 – 09/2026**
Université Paris Dauphine-PSL, Tunis Campus

- Developed AI and autonomous agent features (**Gemini**, **Google Cloud/Vertex AI**) for a mathematics student support platform; integrated a **RAG** system to improve the relevance and personalization of responses.

Mobile Development Intern – Grow Up Technologies: *Comptazone* app (React Native) **06/2024 – 07/2024**
Web Development Intern – STE COMPT-INFO SERVICES: website for an accounting firm (PHP) **06/2023 – 07/2023**

Projects

PostureAI – Real-Time Posture Assistant *(Computer Vision)* **09/2026**

- Local posture-correction assistant using computer vision (**MediaPipe**): head/neck/shoulder tracking, personalized calibration, smart alerts and a tracking dashboard; 100% local processing.
  Python / OpenCV / MediaPipe

AI Platform for Breast Cancer Prediction *(Machine Learning & MLOps)* **05/2026**

- End-to-end MLOps pipeline: experiment tracking (**MLflow**), data drift detection (**Evidently**), **Docker** containerization and **CI/CD**; 6 ML models compared (KNN selected, F1-score 81.00%).
  Python / Scikit-learn / Streamlit / MLflow / Evidently / Docker / GitHub Actions

Drought Risk Forecasting in Tunisia *(Deep Learning)* **05/2026**

- ML/DL pipeline on Tunisian climate data (2017–2025): Random Forest ($\mathbf{R^2 = 0.821}$) then tuned LSTM ($\mathbf{R^2 = 0.953}$); risk mapping by governorate.
  Python / XGBoost / TensorFlow-Keras / Geopandas

Credit Card Fraud Detection *(Data Mining)* **03/2026**

- Highly imbalanced dataset (fraud $< 0.2\%$): SMOTE resampling, supervised classification (**AUC-ROC = 0.98**), clustering and interpretability via SHAP/LIME.
  Python / Scikit-learn / SMOTE / SHAP

BreastFriend – AI Assistant in Oncology *(Generative AI / RAG)* **12/2024**

- Medical RAG chatbot, winner of the **Business Impact Award** (*Hack for Good with Gen AI*): computer vision-based recommendations, patient–doctor follow-up.
  Python / RAG / NLP / Computer Vision

Activities & Awards

- **Vice President, ITLAB Club** (AI Coordinator 2023–25).
- Organizing committee of the *Hack for Good with Gen AI* and *Capture The Flag* hackathons.
- **First Prize, E5t Fund 2024** (environmental association) – *Ocean Wash* project: marine waste collection robot.
- **Third Prize for Innovation**, Spring University 2023.
- Former international judoka – First place, Tunisian Championship.

Languages

Arabic: Native | French: Advanced (C1 - TCF) | English: Advanced (C1) | German: Elementary (A2)